

MultiEURLEX – A multi-lingual and multi-label legal document classification dataset for zero-shot cross-lingual transfer

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Abstract

We introduce MULTI-EURLEX, a new multi-lingual dataset for topic classification of legal documents. The dataset comprises 65k European Union (EU) laws, officially translated in 23 languages, annotated with multiple labels from the EUROVOC taxonomy. We highlight the effect of temporal concept drift and the importance of chronological, instead of random splits. We use the dataset as a testbed for zero-shot cross-lingual transfer, where we exploit annotated training documents in one language (source) to classify documents in another language (target). We find that fine-tuning a multilingually pretrained model (XLM-ROBERTA, MT5) in a single source language leads to catastrophic forgetting of multilingual knowledge and, consequently, poor zero-shot transfer to other languages. Adaptation strategies, namely partial fine-tuning, adapters, BITFIT, LNFIT, originally proposed to accelerate fine-tuning for new end-tasks, help retain multilingual knowledge from pretraining, substantially improving zero-shot cross-lingual transfer, but their impact also depends on the pretrained model used and the size of the label set.

1 Introduction

Multilingual learning is an active field of research in NLP. Starting from neural machine translation (Stahlberg, 2020), multilingual neural models are increasingly being considered across NLP tasks and multilingual benchmark datasets for cross-lingual language understanding are becoming available (Hu et al., 2020; Ruder et al., 2021), complementing previous monolingual benchmarks (Wang et al., 2018). The initial paradigm of multilingual word embeddings (Ruder et al., 2017) was rapidly expanded to pretrained multilingual models (Conneau et al., 2018), including work on zero-shot cross-lingual transfer (Artetxe and Schwenk, 2019). Multilingual models based on TRANSFORMERS (Vaswani et al., 2017), jointly pretrained on large

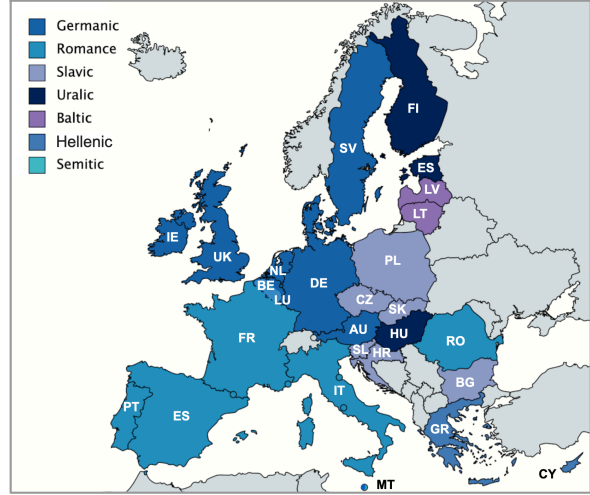


Figure 1: MULTI-EURLEX covers 23 official EU languages (Table 1) from 7 families (illustrated per EU country in the map). The UK was an EU member until 2020. The map should not be taken to imply that no other languages are spoken in EU countries.

corpora across multiple languages, have significantly advanced the state-of-the-art in cross-lingual tasks (Conneau et al., 2020; Xue et al., 2021).

In another interesting direction, legal NLP (Aletas et al., 2019; Zhong et al., 2020) is an emerging field targeting tasks such as legal judgment prediction (Aletas et al., 2016), legal topic classification (Chalkidis et al., 2019), legal question answering (Kim et al., 2015), contract understanding (Hendrycks et al., 2021), to name a few. Generic pretrained language models for legal text in particular have also been introduced (Chalkidis et al., 2020b). But despite rapid growth, cross-lingual transfer has not yet been explored in legal NLP.

To facilitate research on cross-lingual transfer for text classification and legal topic classification in particular, we introduce a new multilingual dataset, MULTI-EURLEX, which includes 65k European Union (EU) laws, officially translated in the 23 EU official languages (Fig. 1). Each document is annotated with multiple labels from EUROVOC, where concepts are organized hierarchi-

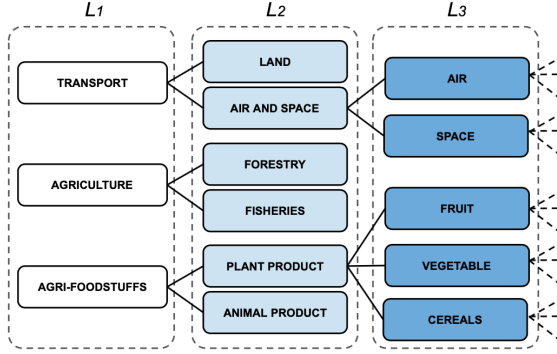


Figure 2: Examples from levels (L_i) 1 to 3 from the EUROVOC hierarchy. More general concepts become more specific as we move from higher to lower levels.

cally (Fig. 2).¹ We use the dataset as a testbed for zero-shot cross-lingual transfer in cases where we wish to exploit labeled training documents in one language (source) to classify documents in another language (target). This would allow, e.g., classifiers trained in resource-rich languages to be reused in languages with fewer or no training instances.

We experiment with monolingual and multilingual TRANSFORMER-based models, i.e., monolingual BERT models (Devlin et al., 2019), XLM-ROBERTA (Conneau et al., 2020), and MT5 (Xue et al., 2021). We find that fine-tuning a multilingual model in a single source language leads to catastrophic forgetting of multilingual knowledge and, consequently, poor zero-shot transfer to target languages. We show that adaptation strategies, namely, not fine-tuning some layers, adapters (Houlsby et al., 2019), BITFIT (Zaken et al., 2021), and LNFIT inspired by Frankle et al. (2021), originally proposed to accelerate fine-tuning for new end-tasks, help retain multilingual knowledge from pretraining, substantially improving zero-shot cross-lingual transfer, but their impact also depends on the particular pretrained model used and the size of the label set. We also compare chronological vs. random splits, highlighting the impact of temporal concept drift in legal topic classification, which causes random splits to over-estimate performance (Søgaard et al., 2021). Our main contributions are:

- A parallel multilingual annotated dataset for legal topic classification with 65k EU laws in 23 languages, which can be used as a testbed for cross-lingual multi-label classification.
- Extensive experiments with state-of-the-art monolingual and multilingual models in 23 languages, which establish strong baselines for research on cross-lingual (legal) text classification.

¹<http://eurovoc.europa.eu/>

- Experiments with several adaptation strategies showing that adaptation is beneficial in zero-shot cross-lingual transfer, apart from task transfer.
- Comparison of chronological vs. random splits, showing the temporal concept drift in legal topic classification and problems with random splits.

2 Related Work

Legal topic classification has been studied for EU legislation (Mencia and Fürnkranz, 2007; Chalkidis et al., 2019) in a monolingual setting (English). While there are several legal NLP studies with non-English datasets (Kim et al., 2015; Walzl et al., 2017; Nguyen et al., 2018; Angelidis et al., 2018; Luz de Araujo et al., 2020), cross-lingual transfer has not been studied in the legal domain.

Cross-lingual transfer is a very active area of wider NLP research, currently dominated by large multilingually pretrained models (Conneau et al., 2018; Eisenschlos et al., 2019; Liu et al., 2020; Xue et al., 2021). Recent work explores adapter modules (Houlsby et al., 2019) to transfer monolingually pretrained (Artetxe et al., 2020) or multilingually pretrained (Pfeiffer et al., 2020) models to new (target) languages. We examine more adaptation strategies, apart from adapter modules, in truly zero-shot cross-lingual transfer. Unlike Pfeiffer et al. (2020), we do not train language-specific adapters per target language; we use adapters to fine-tune a *single* multilingual model on the source language, which is then used in all target languages.

In the broader field of multilingual legal studies, Goncalves and Quaresma (2010) examined legal topic classification with a dataset comprising 2.7k EU laws in 4 languages (English, German, Spanish, Portuguese). They experimented with monolingual SVM classifiers and their combination as a multilingual ensemble. More recently, Galassi et al. (2020) transferred sentence-level gold labels from annotated English to non-annotated German sentences, for the task of identifying unfair clauses in Terms of Service (2.7k sentences) and Privacy Policy documents (1.8k). They experimented with similarity-based methods aligning the English sentences to machine-translations of the German sentences. We experiment with state-of-the-art multilingual TRANSFORMER-based models considering many more languages (23) and a much larger dataset (65k EU laws). Although MULTI-EURLEX is largely parallel, we use it as a testbed for zero-shot cross-lingual transfer, *without* requiring parallel training data or machine translation systems.

Language	ISO code	Member Countries where official	EU Speakers (%)		Number of Documents			Words per document
			Native	Total	Train	Dev.	Test	
English	en	United Kingdom (1973–2020), Ireland (1973), Malta (2004)	13%	51%	55,000	5,000	5,000	1200 / 460
German	de	Germany (1958), Belgium (1958), Luxembourg (1958)	16%	32%	55,000	5,000	5,000	1085 / 410
French	fr	France (1958), Belgium(1958), Luxembourg (1958)	12%	26%	55,000	5,000	5,000	1280 / 480
Italian	it	Italy (1958)	13%	16%	55,000	5,000	5,000	1210 / 460
Spanish	es	Spain (1986)	8%	15%	52,785	5,000	5,000	1380 / 530
Polish	pl	Poland (2004)	8%	9%	23,197	5,000	5,000	1200 / 420
Romanian	ro	Romania (2007)	5%	5%	15,921	5,000	5,000	1500 / 500
Dutch	nl	Netherlands (1958), Belgium (1958)	4%	5%	55,000	5,000	5,000	1230 / 470
Greek	el	Greece (1981), Cyprus (2008)	3%	4%	55,000	5,000	5,000	1230 / 470
Hungarian	hu	Hungary (2004)	3%	3%	22,664	5,000	5,000	1120 / 370
Portuguese	pt	Portugal (1986)	2%	3%	23,188	5,000	5,000	1290 / 500
Czech	cs	Czech Republic (2004)	2%	3%	23,187	5,000	5,000	1170 / 410
Swedish	sv	Sweden (1995)	2%	3%	42,490	5,000	5,000	1130 / 470
Bulgarian	bg	Bulgaria (2007)	2%	2%	15,986	5,000	5,000	1480 / 510
Danish	da	Denmark (1973)	1%	1%	55,000	5,000	5,000	1080 / 410
Finnish	fi	Finland (1995)	1%	1%	42,497	5,000	5,000	890 / 320
Slovak	sk	Slovakia (2004)	1%	1%	15,986	5,000	5,000	1180 / 410
Lithuanian	lt	Lithuania (2004)	1%	1%	23,188	5,000	5,000	1070 / 370
Croatian	hr	Croatia (2013)	1%	1%	7,944	2,500	5,000	1490 / 500
Slovene	sl	Slovenia (2004)	<1%	<1%	23,184	5,000	5,000	1170 / 400
Estonian	et	Estonia (2004)	<1%	<1%	23,126	5,000	5,000	950 / 330
Latvian	lv	Latvia (2004)	<1%	<1%	23,188	5,000	5,000	1080 / 380
Maltese	mt	Malta (2004)	<1%	<1%	17,521	5,000	5,000	1250 / 430

Table 1: MULTI-EURLEX statistics per language: ISO code; EU countries using the language officially (year the country joined the EU in brackets); percentage of EU population speaking the language natively or in total (as native or non-native speakers);³ documents in training, development, test splits; words per document (mean/median).

3 The MULTI-EURLEX Dataset ²

Documents: MULTI-EURLEX comprises 65k EU laws in 23 official EU languages (Table 1). Each EU law has been annotated with EUROVOC concepts (labels) by the Publications Office of EU. Each EUROVOC label ID is associated with a *label descriptor*, e.g., ⟨60, ‘agri-foodstuffs’⟩, ⟨6006, ‘plant product’⟩, ⟨1115, ‘fruit’⟩. The descriptors are also available in the 23 languages. Chalkidis et al. (2019) published a *monolingual* (English) version of this dataset, called EURLEX57K, comprising 57k EU laws with the originally assigned gold labels.

Languages: MULTI-EURLEX covers 23 languages from 7 families (Fig. 1). EU laws are published in all official EU languages, except for Irish for resource-related reasons.³ This wide coverage makes the dataset a valuable testbed for cross-lingual transfer. All languages use the Latin script, except for Bulgarian (Cyrillic script) and Greek.

²The dataset is available at https://huggingface.co/datasets/multi_eurlex. Following Gebru et al. (2018), we provide an extended Dataset Card in Appendix D.

³https://europa.eu/european-union/about-eu/eu-languages_en

⁴Data from European Commission (2012). Following BREXIT (2020), UK citizens are no longer considered EU citizens, thus native English speakers became approx. 1% in the EU, as of 2021. Table 1 includes UK citizens.

Label Set	No. of Labels	In training docs	In all docs
Level 1	21	21 (100%)	21 (100%)
Level 2	127	127 (100%)	127 (100%)
Level 3	567	500 (88%)	511 (90%)
All	7,390	4,220 (57%)	4,591 (62%)

Table 2: EUROVOC concepts in the four label sets and how many are used in the training or entire dataset.

Multi-granular Labeling: EUROVOC has eight levels of concepts (Fig. 2 illustrates three). Each document is assigned one or more concepts (labels). If a document is assigned a concept, the ancestors and descendants of that concept are typically not assigned to the same document. The documents were originally annotated with concepts from levels 3 to 8. We created three alternative sets of labels per document, by replacing each assigned concept by its ancestor from level 1, 2, or 3, respectively. Thus, we provide four sets of gold labels per document, one for each of the first three levels of the hierarchy, plus the original sparse label assignment.⁵ Table 2 presents the distribution of labels across label sets.

Supported Tasks: Similarly to EURLEX57K (Chalkidis et al., 2019), MULTI-EURLEX can be used for legal topic classification, a multi-label classification task where legal documents need to

⁵Levels 4 to 8 cannot be used independently, as many documents have gold concepts from the third level; thus many documents will be mislabeled, if we discard level 3.

be assigned concepts (in our case, from EUROVOC) reflecting their topics. Unlike EURLEX57K, however, MULTI-EURLEX supports labels from three different granularities (EUROVOC levels). More importantly, apart from monolingual (*one-to-one*) experiments, it can be used to study cross-lingual transfer scenarios, including *one-to-many* (systems trained in one language and used in other languages with no training data), and *many-to-one* or *many-to-many* (systems jointly trained in multiple languages and used in one or more other languages).

Data Split and Concept Drift: MULTI-EURLEX is *chronologically* split in training (55k, 1958–2010), development (5k, 2010–2012), test (5k, 2012–2016) subsets, using the English documents. The test subset contains the same 5k documents in all 23 languages (Table 1).⁶ For the official languages of the seven oldest member countries, the same 55k training documents are available; for the other languages, only a subset of the 55k training documents is available (Table 1). Compared to EURLEX57K (Chalkidis et al., 2019), MULTI-EURLEX is not only larger (8k more documents) and multilingual; it is also more challenging, as the chronological split leads to temporal real-world *concept drift* across the training, development, test subsets, i.e., differences in label distribution and phrasing, representing a realistic *temporal generalization* problem (Huang and Paul, 2019; Lazariidou et al., 2021). Recently, Søgaard et al. (2021) showed this setup is more realistic, as it does not over-estimate real performance, contrary to random splits (Gorman and Bedrick, 2019).

Label Set	Random		Chronological	
	<i>train-dev</i>	<i>train-test</i>	<i>train-dev</i>	<i>train-test</i>
Level 1	0.00	0.00	0.03	0.04
Level 2	0.00	0.00	0.12	0.16
Level 3	0.01	0.01	0.21	0.32
All	0.20	0.20	1.09	1.67

Table 3: KL-divergence of label distributions between subsets, using a *random* or *chronological* split.

To verify that the chronological split of MULTI-EURLEX in training, development, test subsets leads to a *temporal concept drift*, we compare the KL-divergence between the label distributions of the subsets using the chronological vs. a random split. Table 3 shows a random split leads to almost zero divergence for levels 1–3 and low divergence

⁶The development subset also contains the same 5k documents in 23 languages, except Croatian. Croatia is the most recent EU member (2013); older laws are gradually translated.

when using all labels. With the chronological split, the divergence increases as the number of labels increases, and is larger between the train and test subsets, which have a larger temporal distance compared to the train and development subsets.

Data Split	Training	Development	Test
Random	99.2	74.7	74.0
Chronological	96.7	58.7	48.4

Table 4: Results of MULTI-EURLEX for the original sparse annotation (7,390 labels) with BERT using a *random* or *chronological* split. Here the model is fine-tuned and tested on English data only (*one-to-one*).

To further highlight the temporal concept drift, we fine-tune BERT (Devlin et al., 2019) on the English part of MULTI-EURLEX using all labels, following Chalkidis et al. (2019). Table 4 shows that although the performance on training data is very high with both splits, it deteriorates more rapidly on development data with the chronological split. Also, performance is stable when moving from development to test data with the random split, since both subsets contain randomly sampled unseen documents; but with the chronological split, performance continues to decline on test data. This confirms our hypothesis of a temporal concept drift and shows that the random split over-estimates real performance, contrary to the chronological split.

4 Methods

4.1 Pretrained Models

NATIVE-BERTS: Many monolingual pretrained TRANSFORMER-based (Vaswani et al., 2017) models have been released, based on BERT (Devlin et al., 2019) or ROBERTA (Liu et al., 2019).⁷ Across classification experiments, L is the cardinality of the label set, and D_h the dimensionality of the hidden states. We feed the top-level hidden state of the $[\text{cls}]$ token ($\in \mathbb{R}^{D_h}$) to a dense layer ($W_{[\text{cls}]} \in \mathbb{R}^{D_h \times L}$) with L outputs and sigmoids.

XLM-ROBERTA: Conneau et al. (2020) introduced a multilingual ROBERTA for 100 languages. It is pretrained on Common Crawl with a vocabulary of 250k sub-words shared across languages. We use the same classification setup as in NATIVE-BERTS.

MT5: Xue et al. (2021) released a multilingual variant of T5 (Raffel et al., 2020), an encoder-decoder TRANSFORMER-based model, pretrained on text in 101 languages from Common Crawl. As in T5, Xue et al. (2021) frame all NLP tasks (incl. text

⁷Appendix C lists the native pretrained BERTs we used.

classification) as text generation. This approach (text-to-text) is reasonable in single-label multi-class classification tasks like those of GLUE (Wang et al., 2018), where the output is expected to be the textual descriptor of a single class. But in our case, we have a *multi-label* task with 5 labels per document on average and label sets containing hundreds or thousands of labels; hence a textual output would be unnecessarily complex. Also, requiring a *sequence* of labels as output would be problematic, since the correct labels are not ordered. Hence, we use only the encoder of MT5. Similarly to XLM-ROBERTA, we add a `[cls]` special token, always at the beginning of the sequence, and use its top-level hidden state to represent the document.⁸

4.2 Cross-lingual Adaptation Strategies

We mainly study *zero-shot cross-lingual transfer*, where we fine-tune (further train) a multilingual model (pretrained on a multilingual corpus) only on annotated documents of a *source* language, and evaluate it (without any further training) on test (and development) documents in the other 22 languages (*one-to-many*). To avoid *catastrophically forgetting* the multilingual pretraining when fine-tuning only for the source language, we examine adaptation strategies, where the model is only partially fine-tuned. These were originally proposed to accelerate fine-tuning when moving to new end-tasks, but we employ them to retain multilingual knowledge. The four strategies are the following:

Frozen layers: In this case, we follow Rosenfeld and Tsotsos (2019) and do not update the parameters of the first N or all ($N = 12$) stacked TRANSFORMER blocks in fine-tuning; we also never update any input embeddings (of tokens, positions, segments). We experiment with $N = 3, 6, 9, 12$.

Adapter modules: In this case, we follow Houlsby et al. (2019), placing adapter modules after each feed-forward layer (FFNN) inside each TRANSFORMER encoder block. Each block contains two FFNN layers: one after the attention layer and one at the very end. An adapter module consists of a down-projection dense layer ($W_{down} \in \mathbb{R}^{D_h \times K}$, assuming row-vectors, where $K \ll D_h$) and a consecutive up-projection ($W_{up} \in \mathbb{R}^{K \times D_h}$), followed

by a residual connection (He et al., 2016). The rest of the Transformer block is not updated, except for layer normalization components (Ba et al., 2016).

BitFit: BITFIT (Zaken et al., 2021) keeps the whole network frozen during fine-tuning, except for bias terms. Zaken et al. showed that applying BITFIT on the English BERT (updating 0.09% of parameters) is competitive with fully fine-tuning the entire model in the GLUE benchmark (Wang et al., 2018).

LNFit: Similarly, Frankle et al. (2021) train only the parameters of *batch normalization* (Ioffe and Szegedy, 2015) layers in image classifiers. We adopt a similar approach, dubbed LNFit, where we fine-tune only the *layer normalization* parameters of pre-trained TRANSFORMERS for text.

The randomly-initialized classification (dense) layer on top of the encoder is always fine-tuned.

5 Experimental Setup

Configuration of Models and Training Details:

We implemented all methods in TENSORFLOW 2, obtaining pretrained models from the Hugging Face library. We release our code and data for reproducibility.⁹ All models follow the BASE configuration with 12 stacked TRANSFORMER encoder blocks, each with $D_h = 768$ and 12 attention heads. We use the Adam optimizer (Kingma and Ba, 2015) across all experiments. We grid-search to tune the learning rate per method, considering classification performance on development data.¹⁰

Evaluation: Given the large number and skewed distribution of labels, retrieval measures have been favored in large-scale multi-label text classification literature (Mullenbach et al., 2018; Chalkidis et al., 2019). Following Chalkidis et al. (2019, 2020a), we report *mean R-Precision* (mRP) (Manning et al., 2009). That is, for each document, the model ranks the labels it selects by decreasing confidence, and we compute $\text{Precision}@k$, where k is the document’s number of gold labels; we then average over documents. For all experiments, we use the chronological data split and report the average across three runs. Unless stated otherwise, we use level 3 with $L = 567$ labels (Table 2), which has a highly skewed (long-tail) label distribution and temporal concept drift (Table 3). In Section 6.2, we also consider label sets from the other levels.

⁸In additional experiments, we also examined the original generative fine-tuning of MT5 (Xue et al., 2021), and another simplified encoder-decoder variant of MT5 agnostic of label order. Both led to worse performance, while being substantially larger (40% more parameters). See Appendix B.

⁹Our code is available on Github (<https://github.com/nlpaueb/multi-eurlex>).

¹⁰See Appendix A for details on hyper-parameter tuning.

	GERMANIC					ROMANCE					SLAVIC			URALIC			
	en	da	de	nl	sv	ro	es	fr	it	pt	pl	bg	cs	hu	fi	el	All
One-to-one (Fine-tune XLM-ROBERTA or monolingually pretrained BERTs in one language, test in the <i>same</i> language.)																	
NATIVE-BERT	67.7	65.5	68.4	66.7	68.5	68.5	67.6	67.4	67.9	67.4	67.2	-	66.7	67.7	67.8	67.8	67.4
XLM-ROBERTA	67.4	66.7	67.5	67.3	66.5	66.4	67.8	67.2	67.4	67.0	65.0	66.1	66.7	65.5	66.5	65.8	66.6
Diff.	-0.3	+1.2	-0.9	+0.6	-2.0	-2.1	+0.2	-0.2	-0.5	-0.4	-2.2	-	0.0	-2.2	-1.3	-2.0	-0.7
One-to-many (Fine-tune XLM-ROBERTA <i>only</i> in English, test in all languages, with alternative adaptation strategies.)																	
End-to-end fine-tuning	67.4	56.5	52.4	49.0	55.7	55.2	54.0	55.0	52.0	50.5	46.9	51.2	49.6	48.8	46.4	33.3	49.3
First 3 blocks frozen	66.3	59.1	56.8	55.3	57.5	57.9	58.1	57.7	56.2	54.9	53.7	56.1	54.3	51.0	52.1	42.4	53.0
First 6 blocks frozen	66.3	59.1	57.4	55.7	57.9	57.2	56.9	57.9	53.9	55.4	51.9	55.8	52.6	47.3	48.7	39.6	51.7
First 9 blocks frozen	65.8	59.4	57.9	56.9	58.6	58.2	58.7	59.4	55.7	57.5	53.4	56.7	54.2	48.8	50.4	44.5	53.0
All 12 blocks frozen	27.2	21.4	24.6	24.6	23.0	21.6	23.4	21.9	20.1	25.1	22.8	23.1	24.3	22.8	21.9	19.0	22.2
Adapter modules	67.3	61.5	59.3	57.8	59.5	60.3	61.0	60.4	58.8	58.5	57.5	59.2	56.8	55.3	55.6	46.1	56.1
BITFIT (bias terms only)	63.9	59.3	57.0	54.0	58.2	57.8	57.4	56.9	56.4	55.5	54.0	55.6	54.8	51.2	54.8	42.1	53.7
LNFIT (layer-norm only)	63.1	58.9	55.7	54.1	56.6	59.1	59.1	58.0	56.6	57.2	55.7	55.4	52.8	51.4	50.7	39.9	53.3
Many-to-many (Jointly fine-tune XLM-ROBERTA in <i>all</i> languages, test in all languages, with alternative adaptation strategies.)																	
End-to-end fine-tuning	66.4	66.2	66.2	66.1	66.1	66.3	66.3	66.2	66.3	65.9	65.6	65.7	65.7	65.2	65.8	65.1	65.7
Adapter modules	67.2	67.1	66.3	67.1	67.0	67.4	67.2	67.1	67.4	67.0	66.2	66.6	67.0	65.5	66.6	65.7	66.4

Table 5: Test results for level 3 (567 labels) of MULTI-EURLEX. We show mRP (%) for the 16 most widely spoken EU official languages, and mRP averaged over all 23 languages. Appendix E reports results for all languages.

6 Experiments and Discussion

For the main experiments, we mainly use XLM-ROBERTA in a *one-to-many* setting (fine-tuning in English, testing in all languages). We also report key MT5 results for completeness. As a ceiling for cross-lingual transfer, in Section 6.1 we first evaluate monolingual (native) BERT models and XLM-ROBERTA, both in a *one-to-one* manner (fine-tuning and testing in the same language), which requires annotated training data in the target language. For completeness, in Section 6.3, we also report *many-to-many* results, where XLM-ROBERTA is jointly fine-tuned and tested in all languages.

6.1 Monolingual Classification (*one-to-one*)

Table 5 (top) shows that in the *one-to-one* setting, XLM-ROBERTA is competitive to native (monolingually pretrained) BERTs with a minor decrease of 0.7 mRP on average across languages. Of course, the *one-to-one* setting requires training data in the target language. We report these results as an *upper bound* for zero-shot cross-lingual transfer. Also, the native BERTs are pretrained on corpora of different sizes and quality, which explains why they are not consistently better than XLM-ROBERTA.

6.2 Cross-lingual Transfer (*one-to-many*)

XLM-ROBERTA adaptation: In the *one-to-many* setting, where we fine-tune in English and test in all languages, Table 5 (middle) shows that all adaptation strategies vastly improve the performance of XLM-ROBERTA across languages (up to 6.8 All mRP increase) comparing to no adaptation (end-to-end fine-tuning), while remaining competitive in English (source). This indicates that not fine-tuning the full set of parameters helps the model

retain more of its multilingual knowledge obtained during pretraining. We observe no big difference among the block freezing strategies for $N = 3, 6, 9$, but performance deteriorates substantially when all blocks are frozen ($N = 12$).¹¹ We speculate there is a trade-off between freezing more blocks to retain multilingual knowledge and freezing fewer blocks to benefit end-task (classification) performance. Adapters consistently lead to the best results in all languages and overall (All mRP 56.1), with practically no decrease in English (source) performance (67.3). BITFIT, which only fine-tunes bias terms (4e-2% of parameters), and LNFIT, which fine-tunes even fewer parameters (1e-2%), are the second and third best strategies. These results highlight the expressive power of the few parameters BITFIT and LNFIT modify; this observation has been also discussed in previous studies (Frankle et al., 2021; Zaken et al., 2021), but not in a multi-lingual setting. Overall, fine-tuning in a single language leads to substantial forgetting of multilingual knowledge, but adaptation strategies, especially adapter modules, alleviate this problem and improve cross-lingual end-task performance.

MT5 adaptation: In Table 6, we repeat the *one-to-many* experiments of Table 5, this time with MT5 (encoder only). For brevity, we report only mRP on the source (English) language, and mRP averaged over the 23 languages.¹² BITFIT cannot be applied in this case, because MT5 does not use bias terms. As in Table 5, freezing the initial N blocks of the encoder ($N = 3, 6, 9$) improves cross-lingual transfer (average mRP increase up to 4.7), but freezing

¹¹When $N = 12$, we practically evaluate (probe) the intact pre-training knowledge of XLM-ROBERTA in the end-task.

¹²See Appendix E for additional experimental results.

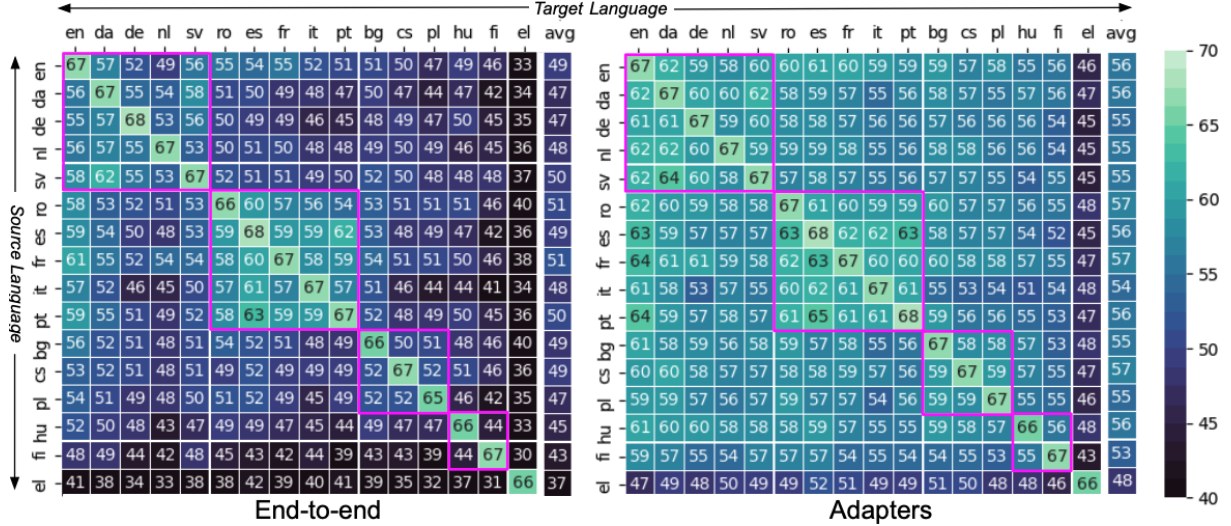


Figure 3: Test results (mRP, %) for Level 3 (567 labels) with XLM-ROBERTA, when fine-tuning in one language (source, rows) and testing in all languages (columns), without adaptation (end-to-end, left) and with adapter modules (right). The languages are grouped (framed) in language families (Germanic, Romance, Slavic, Uralic).

Adaptation strategy	Params (%)	en (Src)	All
End-to-end fine-tuning	277M (100.0%)	67.4	53.7
First 3 blocks frozen	63.7M (23.0%)	67.4	56.9
First 6 blocks frozen	42.4M (15.3%)	66.3	58.4
First 9 blocks frozen	21.2M (7.7%)	68.0	58.3
All 12 blocks frozen	— (0.0%)	20.2	16.8
Adapter modules	7.1M (1.7%)	66.3	44.0
LNFIT (layer-norm only)	19.2K (0.01%)	59.5	38.7

Table 6: Test results of MT5 fine-tuned in English (en). We show mRP (%) in English (Src), and averaged across all 23 languages (All). We also report the trainable parameters (excl. the classification layer).

all layers ($N = 12$) harms performance. Surprisingly adapter modules, which are the best adaptation strategy for XLM-ROBERTA (Table 5, middle), lead to very poor performance (average mRP 44); there are similar results with LNFIT (average mRP 38.7). We speculate this happens because the encoder of MT5 needs to ‘re-program’ itself during fine-tuning to perform as a stand-alone encoder; in adapters and LNFIT ‘re-programming’ is only facilitated by very few parameters and the model is ‘forced’ (due to low adaptable capacity) to discard multilingual knowledge aggressively. XLM-ROBERTA follows the opposite pattern (fewer parameters lead to better cross-lingual transfer), because it is pre-trained as a stand-alone encoder. We leave a more thorough investigation of the trade-off between the number of trainable parameters vs. end-task (Src/All) performance for future work.

Different source languages: In the cross-lingual experiments so far, we fine-tuned the model in English (source) and evaluated it in all 23 languages. In Fig. 3, we repeat these experiments using a *different*

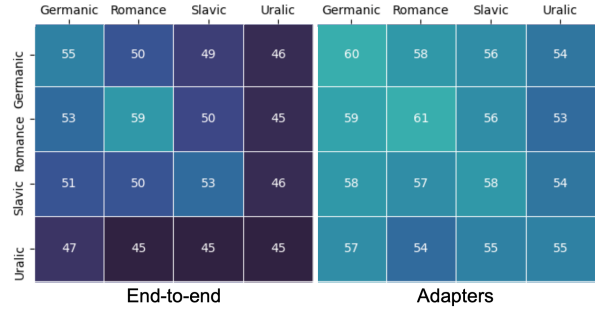


Figure 4: Cross-lingual test results (mRP, %) for level 3 (567 labels) with XLM-ROBERTA, averaged over language families (transfer from one family to another).

different source language in each repetition (rows), evaluating again in all languages (columns).¹³ We use XLM-ROBERTA without adaptation (end-to-end, left) or with adapter modules (right). Despite the dominance of English in multi-lingual NLP literature, we observe that using alternative source languages (e.g., Romanian or French) lead to better target results. Similar results have been presented in Turc et al. (2021) for other NLP tasks. As in the previous one-to-many experiments with XLM-ROBERTA (Table 5, middle), adapters vastly improve cross-lingual transfer across all cases (e.g., English-en to Danish-da improves from 57 to 62 mRP), with occasionally slightly lower monolingual performance (e.g., German-de drops from 68 to 67). Cross-lingual transfer performs overall better when the source and target languages are in the same family (frames of Fig. 3), especially for Romance languages (Fig. 4, diagonal).¹³ Also, when using adapters, cross-lingual performance often

¹³See Appendix E for additional experimental results.

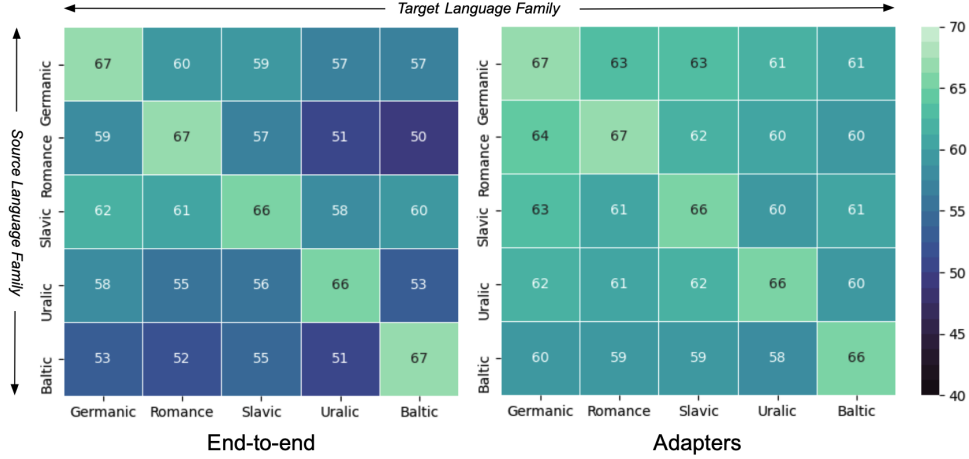


Figure 5: Cross-lingual test results (mRP, %) for level 3 (567 labels) with XLM-ROBERTA, when fine-tuning *end-to-end* or with adapters in *all languages of the same family* (Src) and testing is averaged over each language family.

drops less abruptly when moving outside of the family of the source language. For example, when fine-tuning in Danish-da, if the test set changes from Swedish-sv to Spanish-es, performance drops from 58 to 50 without adapters (Fig. 3, left), but the change is smoother, from 62 to 59, with adapters (Fig. 3, right). This is better illustrated in the right part of Fig. 4 (smoother changes across cells per row). These results confirm that adapter modules help retain more multilingual knowledge.

Transfer from one family to another: In the previous experiment (“*Different source languages*”), we used a *different source language in each repetition* and evaluated in all languages. To better understand how linguistic proximity between families affects performance, in Figure 5 we present additional experiments in a *many-to-many* setting, where each model is trained across *all languages in the same family* (source) and evaluated across all languages. We use again XLM-ROBERTA without adaptation (end-to-end, left) or with adapter modules (right). We observe (Fig. 5, left) that cross-lingual transfer performs overall better when the source and target families are the same. Also, when using adapters (Fig. 5, right), cross-lingual performance drops less abruptly when moving to another family, different from the one (source) whose languages were used for fine-tuning. As expected, the cross-lingual performance of these models (jointly fine-tuned in a language family) is substantially higher than the ones trained in a *one-to-one* setting (Fig. 3–4), and closer to that of the models jointly fine-tuned in *all 23 languages* (*many-to-many*, results reported in the lower part of Table 5).

Version of the input text	en (Src)		Rest	
	<i>T</i>	mRP	<i>T</i> (%)	mRP
Full-text	100%	67.3	100%	56.1
w/o digits	89%	67.1	88%	55.0
w/o digits & English vocab.	22%	14.0	77%	51.5

Table 7: Test results of XLM-ROBERTA (with adapters) removing digits and words used in the English part of MULTI-EURLEX during inference. We show mRP (%) for English (Src) and averaged over the other 22 languages (Rest). *T* is the percentage of tokens retained.

Removing digits and shared words: In an ablation study, during inference we remove digits and words that are shared across languages to see to what extent label predictions depend on them. Initially, we eliminate digits, which constitute approx. 10% of the average document length measured in white-space separated tokens. Digits often participate in legal references (e.g., “*established by Regulation No 1468/81*”) or other coding schemes that may hint EUROVOC concepts (e.g., when specific laws are highly cited). Moreover, inspecting training documents, we observe that vocabulary words (e.g., of Latin origin) are shared to a substantial degree (23% on average) across languages; thus as a second step we remove approx. 25k words used more than 25 times in English documents to break direct cross-lingual alignment. Table 7 shows that removing digits leads to a small decrease in *one-to-one* performance (-0.2) and a larger, though still small, decrease in *one-to-many* performance (-1.1). Eliminating shared words (present in the English vocabulary) leads to a further decrease (-3.5) in cross-lingual performance, and English performance of course plunges as the remaining text is very short and severely corrupted.

Adaptation Strategy	Parameters	Level 1 (21)		Level 2 (127)		Level 3 (567)		Original (7,390)	
		en (Src)	All	en (Src)	All	en (Src)	All	en (Src)	All
End-to-end fine-tuning	278M (100%)	83.2	75.7	73.6	58.7	67.4	49.3	47.6	27.6
First 3 blocks frozen	63.8M (23.0%)	82.9	76.4	71.3	60.2	66.3	53.0	47.3	29.0
First 6 blocks frozen	42.5M (15.3%)	82.3	76.7	69.6	61.1	66.3	51.7	47.1	30.1
First 9 blocks frozen	21.3M (7.7%)	82.0	74.8	70.7	60.1	65.8	53.0	48.0	32.8
Adapter modules	9.5M (3.3%)	83.1	77.2	72.3	61.2	67.3	56.1	47.9	35.1
BITFIT (bias terms only)	101K (0.04%)	82.7	76.1	70.2	60.1	63.9	53.7	48.3	33.9
LNFIT (layer-norm only)	36.8K (0.01%)	81.5	74.9	69.7	59.3	63.1	53.3	43.1	26.4
↑ <i>Averaged Adapt.</i> ↑	-	82.4	76.0	70.6	60.3	65.5	53.5	47.0	31.2
All 12 blocks frozen	- (0.0%)	61.4	56.5	39.0	31.6	27.2	22.2	26.1	15.3

Table 8: Test results of XLM-ROBERTA fine-tuned in English, for all adaptation strategies and different label granularities (EUROVOC levels, Table 2). We show mRP results (%) for English (Src) and averaged over all 23 languages (All). We also count the trainable parameters, excl. the classification layer, which remains the same.

Different label granularities: Table 8 shows XLM-ROBERTA results with labels from different EUROVOC levels (Table 2) for all adaptation strategies. As expected, performance deteriorates (approx. 5-10% per level) as the size of the label set increases. Nonetheless, we observe consistent improvements with adaptation strategies compared to full (end-to-end) fine-tuning for all label sets, with the exception of the fully (all 12 blocks) frozen model (last row). Adapters have the best overall performance, but the ranking and impact of the different adaptation strategies varies across levels. Specifically, as the size of the label set increases, the average (Table 8, second-to-last line) adaptation zero-shot (All) performance: (a) improves compared to no adaptation (end-to-end fine-tuning), approx. $+0.3 \rightarrow +1.6 \rightarrow +4.2 \rightarrow +3.6$, as we move from level 1 to the full (original) label set, with a small drop from level 3 to the full label set; and (b) deteriorates more aggressively when comparing it to English (Src) performance, approx. $-6.4 \rightarrow -10.3 \rightarrow -12.0 \rightarrow -15.8$. The latter (b) is due to the need to model increasingly finer concepts (labels), which complicates cross-lingual concept alignment and, hence, hurts transfer, leaving more scope for adaptation strategies to make a difference (a).

6.3 Multilingual Fine-tuning (*many-to-many*)

In the lower part of Table 5, we report results for XLM-ROBERTA, fine-tuned end-to-end or using adapters, when the model is *jointly fine-tuned in all languages*. In this case, for each epoch and batch we randomly select a language of the document, among the available ones; not all documents are available in all 23 languages (Table 1). Adapter modules again consistently improve performance. The *many-to-many* models largely outperform the *one-to-many* models (Table 5, middle), as they have access to annotated training documents in all languages. Nevertheless, this is still an interesting sce-

nario, because it allows *deploying a single model* that handles all languages and is competitive to using multiple native BERT models, one per language

7 Conclusions and Future Work

We introduced MULTI-EURLEX, a new multilingual legal topic classification dataset with 65k documents (EU laws) in 23 languages, where each document is annotated with multiple labels (concepts) from the EUROVOC taxonomy, with alternative label granularities. To the best of our knowledge, this is one of the most diverse, in terms of languages, classification datasets. We mainly used the dataset as a testbed for zero-shot cross-lingual transfer.

Experimental results showed that fine-tuning a multilingually pretrained model (XLM-ROBERTA, MT5) in a single language leads to catastrophic forgetting of multilingual knowledge and, consequently, poor zero-shot transfer. We found that adaptation strategies, originally proposed to accelerate fine-tuning for end-tasks, help retain multilingual knowledge from pretraining, substantially improving zero-shot cross-lingual transfer. However, their impact depends on the size of the label set, i.e., the gains increase as the label set increases. Interestingly, even adaptation strategies (BITFIT, LNFIT) that fine-tune a very small fraction of parameters ($<0.05\%$) are competitive. Experimental results also showed that multilingual models are competitive to monolingual models in the one-to-one set-up; and that a single multilingual model jointly fine-tuned in all languages is also competitive. We also used MULTI-EURLEX to highlight the effect of temporal concept drift and the importance of chronological, instead of random, splits.

In future, we would like to examine alternative cross-lingual adaptation strategies (Pfeiffer et al., 2020, 2021) and distributionally robust optimization techniques (Sagawa et al., 2020; Koh et al., 2021) to address the temporal concept drift.